Supplementary material

Supplementary Table 1. Multivariable associations of covariates with absolute CSF Aβ40 and total tau across disease stages.

(**A**) Multivariable linear regression models assessing the association of demographic and anatomical covariates (sex, age, intracranial volume, grey matter volume, and ventricular volume) with absolute CSF Aβ40 and total tau levels within each disease group.

	NC		Aβ- MC		Aβ+ aMC		Aβ+ sMC	
	B	*P*-value	B	*P*-value	B	*P*-value	B	*P*-value
CSF Aβ40								
Sex	-150.93	0.79	-323.01	0.64	528.27	0.56	-90.06	0.89
Age	114.85	3.4×10^{-5}***	10.28	0.82	61.72	0.16	113.14	2.6×10^{-4}***
Intracranial volume	3.90×10^{-3}	0.19	-6.40×10^{-4}	0.86	2.50×10^{-3}	0.59	-3.20×10^{-3}	0.23
Grey matter volume	2.50×10^{-4}	0.98	5.20×10^{-3}	0.61	5.30×10^{-3}	0.64	1.80×10^{-2}	0.01**
Ventricular volume	-0.12	2.4×10^{-4}***	-0.09	0.01*	-0.13	0.01*	-3.50×10^{-2}	0.04*
Total tau								
Sex	-9.00×10^{-3}	0.93	0.08	0.58	0.52	0.09	-0.32	0.56
Age	2.10×10^{-2}	1.9×10^{-5}***	-2.20×10^{-2}	0.01*	4.80×10^{-4}	0.97	-7.50×10^{-3}	0.75
Intracranial volume	7.10×10^{-7}	0.20	1.90×10^{-7}	0.8	2.50×10^{-6}	0.11	1.80×10^{-6}	0.39
Grey matter volume	5.40×10^{-7}	0.74	-9.40×10^{-8}	0.96	-5.30×10^{-6}	0.15	-6.80×10^{-6}	0.23
Ventricular volume	-2.30×10^{-5}	1.0×10^{-4}***	-1.80×10^{-5}	0.01*	-5.10×10^{-6}	0.76	-8.70×10^{-6}	0.54

Groups are defined as NC, amyloid-negative non-carriers; Aβ- MC, amyloid-negative mutation carriers; Aβ+ aMC, amyloid-positive asymptomatic mutation carriers with CDR = 0; and Aβ+ sMC, amyloid-positive symptomatic mutation carriers with CDR > 0. Sample sizes were: Aβ40 (n = 167, 88, 89, 96) and total tau (n = 166, 96, 84, 92) across the respective groups. B = unstandardized regression coefficient.

(**B**) Multivariable linear regression models assessing the association of demographic and anatomical covariates *including gene group* with absolute CSF Aβ40 and total tau levels within each disease group.

	Aβ- MC		Aβ+ aMC		Aβ+ sMC	
	B	*P*-value	B	*P*-value	B	*P*-value
CSF Aβ40						
Sex	-287.74	0.68	594.11	0.51	-62.79	0.93
Age	12.21	0.79	62.73	0.16	115.63	2.7×10^{-4}***
Intracranial volume	-8.70×10^{-4}	0.81	2.80×10^{-3}	0.55	-3.20×10^{-3}	0.24
Grey matter volume	5.90×10^{-3}	0.57	3.70×10^{-3}	0.75	1.80×10^{-2}	0.01*
Ventricular volume	-0.08	0.02*	-0.13	0.02*	-3.50×10^{-2}	0.05*
Gene group: *PSEN1*	reference					
Gene group: *PSEN2*	482.39	0.61	1025.3	0.33	-72.88	0.97
Gene group: *APP*	240.9	0.70	-1208.77	0.38	-362.15	0.59
Total tau						
Sex	0.09	0.54	0.49	0.12	-0.3	0.59
Age	-2.10×10^{-2}	0.02*	2.00×10^{-3}	0.90	-4.70×10^{-3}	0.85
Intracranial volume	1.10×10^{-7}	0.88	2.70×10^{-6}	0.09	2.00×10^{-6}	0.36
Grey matter volume	1.30×10^{-7}	0.95	-6.00×10^{-6}	0.11	-6.90×10^{-6}	0.24
Ventricular volume	-1.80×10^{-5}	0.01*	-5.90×10^{-6}	0.73	-8.90×10^{-6}	0.53
Gene group: *PSEN1*	reference					
Gene group: *PSEN2*	0.13	0.47	0.09	0.81	-0.59	0.68
Gene group: *APP*	0.08	0.53	-0.42	0.32	-0.28	0.63

Additional adjustment for mutation type (*APP, PSEN1, PSEN2*) did not materially alter the results and was not significantly associated with biomarker levels. Groups are defined as Aβ- MC, amyloid-negative mutation carriers; Aβ+ aMC, amyloid-positive asymptomatic mutation carriers with CDR = 0; and Aβ+ sMC, amyloid-positive symptomatic mutation carriers with CDR > 0. Gene group was not available in non-carriers. Sample sizes were: Aβ40 (n = 167, 88, 89, 96) and total tau (n = 166, 96, 84, 92) across the respective groups. B = unstandardized regression coefficient.

Supplementary Table 2. Multivariable associations of covariates with the relationship of CSF Aβ40 and total tau across disease stages.

	CSF total tau	Sex	Age	Intracranial volume	Grey matter volume	Ventricular volume
				NC (*n*=166)		
B	4.7e+03	-1.0e+02	6.4e+00	6.2e-04	-2.9e-03	-7.0e-03
P-value	4.3e-47***	0.72	0.66	0.69	0.53	0.68
				Aβ- MC (*n*=106)		
B	3.5e+03	-7.7e+02	5.7e+01	1.7e-03	4.2e-04	-3.6e-02
P-value	8.6e-15***	0.11	0.07	0.53	0.95	0.17
				Aβ+ aMC (*n*=99)		
B	1.3e+03	-7.2e+02	8.8e+01	3.5e-04	8.2e-03	-1.3e-01
P-value	7.4e-08***	0.31	0.01*	0.92	0.33	2.8e-04***
				Aβ+ sMC (*n*=94)		
B	6.0e+02	-2.9e+02	9.7e+01	-4.8e-03	2.1e-02	-3.3e-02
P-value	3.3e-06***	0.63	3.7e-04***	0.05*	1.0e-03***	0.04*

Multivariable linear regression models evaluating the association between CSF Aβ40 and total tau within each disease group, adjusted for the same covariates as Supplementary Table 1. β-coefficients represent effect sizes from models including all covariates simultaneously. Groups are defined as NC, amyloid-negative non-carriers; Aβ- MC, amyloid-negative mutation carriers; Aβ+ aMC, amyloid-positive asymptomatic mutation carriers with CDR = 0; and Aβ+ sMC, amyloid-positive symptomatic mutation carriers with CDR > 0. B = unstandardized regression coefficient. Significant p-values are given * <0.05, ** < 0.01, ***<0.001.

Supplementary Table 3. Regression and group-comparison models of CSF Aβ40, total tau, and their association across disease groups.
(**A**) Linear regression model with CSF total tau as outcome and CSF Aβ40, disease group (NC, Aβ- MC, Aβ+ aMC, Aβ+ sMC), and their interaction as predictors. Using *emtrends*, we estimated the CSF Aβ40 slope within each group and compared slopes pairwise (unadjusted p-values). Using *emmeans* at CSF Aβ40 = 0 (non-carrier reference level), we estimated group-specific intercepts and tested pairwise intercept differences versus non-carriers. Groups are defined as NC, amyloid-negative non-carriers; Aβ- MC, amyloid-negative mutation carriers; Aβ+ aMC, amyloid-positive asymptomatic mutation carriers with CDR = 0; and Aβ+ sMC, amyloid-positive symptomatic mutation carriers with CDR > 0.

Term	Group	mean	SD	*P*
Intercept	NC	0.499	0.151	9.72×10^{-4}***
	Aβ- MC	0.589	0.245	0.017
	Aβ+ aMC	0.435	0.264	0.100
	Aβ+ sMC	2.079	0.248	2.03×10^{-16}***
Slope	NC (reference)	1.66×10^{-4}	1.70×10^{-5}	1.30×10^{-21}***
	Aβ- MC * CSF Aβ40	-5.36×10^{-5}	2.74×10^{-5}	0.052
	Aβ+ aMC * CSF Aβ40	5.29×10^{-5}	2.96×10^{-5}	0.073
	Aβ+ sMC * CSF Aβ40	6.93×10^{-5}	2.99×10^{-5}	0.021*

(**B**) Group differences in mean CSF Aβ40 and mean CSF total tau between disease groups and non-carriers, assessed with t-tests. The table reports group means, standard deviations, and unadjusted p-values for comparisons versus non-carriers.

Biomarker	Group	Mean (pg/mL)	SD	*P* compared to reference
CSF Aβ40	NC	8554	3146	reference
	Aβ- MC	8007	2679	0.243
	Aβ+ aMC	8014	3237	0.233
	Aβ+ sMC	6872	2304	2.04×10^{-4} ***
CSF total tau	NC	1.97	0.58	reference
	Aβ- MC	2.05	0.47	0.724
	Aβ+ aMC	2.62	0.9	0.001**
	Aβ+ sMC	4.67	1.81	3.94×10^{-36} ***

Abbreviations: Aβ = amyloid-beta; CDR= Clinical Dementia Rating; SD = Standard Deviation.

Supplementary Table 4. McNemar test of significant difference in individuals with positive rate of change.

Tau marker	Measure	NC (n = 96)	*P*	Aβ- MC (*n* = 58)	*P*	Aβ+ aMC (*n* = 51[i])	*P*	Aβ+ sMC (*n* = 51)	*P*
p-tau217	**Levels**	**32 (33.3%)**		**34 (58.6%)**		**41 (80.4%)**		**30 (58.8%)**	
	Ratio to Aβ40	35 (36.5%)		38 (65.5%)		46 (90.2%)		43 (84.3%)	
	Difference	3 (3.1%)	0.689	4 (6.9%)	0.423	5 (9.8%)	0.131	13 (25.5%)	1.95e-03**
	Ratio to t-tau	23 (24.0%)		30 (51.7%)		43 (84.3%)		23 (45.1%)	
	Difference	-9 (-9.4%)	0.0809	-4 (-6.9%)	0.343	2 (3.9%)	0.617	-7 (-13.7%)	0.146
p-tau181	**Levels**	**42 (43.8%)**		**26 (44.8%)**		**41 (80.4%)**		**22 (43.1%)**	
	Ratio to Aβ40	52 (54.2%)		44 (75.9%)		42 (82.4%)		38 (74.5%)	
	Difference	10 (10.4%)	0.155	18 (31.0%)	5.2e-04***	1 (2.0%)	1	16 (31.4%)	4.07e-04***
	Ratio to t-tau	39 (40.6%)		31 (53.4%)		36 (70.6%)		17 (33.3%)	
	Difference	-3 (-3.1%)	0.710	5 (8.6%)	0.228	-5 (-9.8%)	0.228	-5 (-9.8%)	0.267
Total tau	**Levels**	**41 (42.7%)**		**31 (53.4%)**		**41 (78.8%)**		**33 (64.7%)**	
	Ratio to Aβ40	50 (52.1%)		44 (75.9%)		42 (80.8%)		45 (88.2%)	
	Difference	9 (9.4%)	0.233	13 (22.4%)	0.0123*	1 (1.9%)	1	12 (23.5%)	9.52e-03**
p-tau205	**Levels**	**33 (34.4%)**		**29 (50.0%)**		**44 (86.3%)**		**32 (62.7%)**	
	Ratio to Aβ40	39 (40.6%)		34 (58.6%)		43 (84.3%)		42 (82.4%)	
	Difference	6 (6.2%)	0.264	5 (8.6%)	0.302	-1 (-2.0%)	1	10 (19.6%)	9.37e-03**
	Ratio to t-tau	37 (38.5%)		27 (46.6%)		37 (72.5%)		34 (66.7%)	
	Difference	4 (4.2%)	0.423	-2 (-3.4%)	0.724	-7 (-13.7%)	0.0704	2 (3.9%)	0.773

For each biomarker, we summarized cross-sectional differences in the proportion of subjects with a positive annual rate of change across carrier groups for the raw levels, ratio to Aβ40, and ratio to total tau. Groups are defined as NC, amyloid-negative non-carriers; Aβ- MC, amyloid-negative mutation carriers; Aβ+ aMC, amyloid-positive asymptomatic mutation carriers with CDR = 0; and Aβ+ sMC, amyloid-positive symptomatic mutation carriers with CDR > 0. Within each carrier group, we classified subjects as "positive" if their estimated raw or ratio slope was >0 and calculated the number and percentage of positives. We then computed the change in the number and percentage of positives (Δn and Δ%) after moving from raw to ratio values. Within each carrier group and biomarker, we used McNemar's test to obtain a p-value for the paired change in positivity status between the raw and ratio classifications. (i) n for Total tau is *n*=52.

Supplementary Table 5. Spearman correlations across CSF biomarkers, raw and Aβ40-normalized, across disease stages.

Disease stage	Predictor	ρ [95% CI]		Δρ	*n*
		Raw levels	Δρ		
Non-carriers	p-tau217 ~ p-tau181	0.86 [0.81-0.9]	0.66 [0.55-0.75]	-0.2	169
	p-tau217 ~ Total tau	0.85 [0.79-0.89]	0.64 [0.54-0.73]	-0.2	169
	p-tau181 ~ Total tau	0.96 [0.94-0.97]	0.88 [0.8-0.92]	-0.09	169
	p-tau217 ~ p-tau205	0.73 [0.65-0.8]	0.49 [0.35-0.61]	-0.25	169
	p-tau181 ~ p-tau205	0.74 [0.65-0.81]	0.58 [0.46-0.69]	-0.16	169
	Total tau ~ p-tau205	0.72 [0.63-0.8]	0.53 [0.41-0.64]	-0.19	169
	p-tau217 ~ NfL	0.47 [0.33-0.59]	0.19 [0.03-0.34]	-0.28	167
	p-tau181 ~ NfL	0.51 [0.37-0.62]	0.22 [0.07-0.35]	-0.29	167
	Total tau ~ NfL	0.49 [0.35-0.6]	0.13 [-0.02-0.28]	-0.35	167
	p-tau205 ~ NfL	0.5 [0.36-0.61]	0.25 [0.09-0.4]	-0.25	167
	p-tau217 ~ YKL-40	0.6 [0.48-0.69]	0.36 [0.23-0.49]	-0.24	160
	p-tau181 ~ YKL-40	0.65 [0.54-0.73]	0.4 [0.26-0.53]	-0.25	160
	Total tau ~ YKL-40	0.62 [0.51-0.72]	0.33 [0.18-0.47]	-0.29	160
	p-tau205 ~ YKL-40	0.51 [0.37-0.62]	0.25 [0.09-0.4]	-0.26	160
	NfL ~ YKL-40	0.68 [0.57-0.78]	0.58 [0.45-0.69]	-0.1	159
Aβ- MC	p-tau217 ~ p-tau181	0.74 [0.63-0.83]	0.71 [0.57-0.81]	-0.03	100
	p-tau217 ~ Total tau	0.64 [0.5-0.77]	0.65 [0.48-0.77]	0	100
	p-tau181 ~ Total tau	0.95 [0.91-0.97]	0.92 [0.86-0.96]	-0.03	101
	p-tau217 ~ p-tau205	0.59 [0.46-0.71]	0.55 [0.38-0.69]	-0.04	100
	p-tau181 ~ p-tau205	0.6 [0.46-0.71]	0.54 [0.35-0.69]	-0.06	101
	Total tau ~ p-tau205	0.56 [0.41-0.69]	0.49 [0.3-0.66]	-0.07	101
	p-tau217 ~ NfL	0.34 [0.13-0.51]	0.35 [0.14-0.54]	0.01	93
	p-tau181 ~ NfL	0.33 [0.13-0.51]	0.41 [0.22-0.58]	0.08	93
	Total tau ~ NfL	0.34 [0.14-0.52]	0.38 [0.18-0.55]	0.04	93
	p-tau205 ~ NfL	0.39 [0.19-0.57]	0.42 [0.2-0.6]	0.02	93
	p-tau217 ~ YKL-40	0.26 [0.03-0.46]	0.31 [0.09-0.52]	0.05	90
	p-tau181 ~ YKL-40	0.4 [0.18-0.59]	0.52 [0.32-0.68]	0.12	90
	Total tau ~ YKL-40	0.41 [0.19-0.59]	0.51 [0.31-0.67]	0.1	90
	p-tau205 ~ YKL-40	0.33 [0.12-0.51]	0.44 [0.23-0.62]	0.11	90
	NfL ~ YKL-40	0.46 [0.26-0.62]	0.53 [0.33-0.68]	0.07	89
Aβ+ aMC	p-tau217 ~ p-tau181	0.97 [0.95-0.98]	0.96 [0.93-0.98]	-0.01	105
	p-tau217 ~ Total tau	0.89 [0.83-0.92]	0.87 [0.8-0.91]	-0.02	105

	p-tau181 ~ Total tau	0.94 [0.91-0.96]	0.93 [0.89-0.96]	-0.01	105
	p-tau217 ~ p-tau205	0.92 [0.88-0.95]	0.89 [0.82-0.92]	-0.04	105
	p-tau181 ~ p-tau205	0.92 [0.88-0.95]	0.91 [0.86-0.94]	-0.02	105
	Total tau ~ p-tau205	0.87 [0.8-0.91]	0.87 [0.8-0.91]	0	105
	p-tau217 ~ NfL	0.71 [0.59-0.79]	0.63 [0.49-0.74]	-0.08	101
	p-tau181 ~ NfL	0.71 [0.6-0.79]	0.65 [0.5-0.76]	-0.06	101
	Total tau ~ NfL	0.6 [0.46-0.72]	0.58 [0.43-0.71]	-0.02	101
	p-tau205 ~ NfL	0.74 [0.62-0.82]	0.7 [0.56-0.8]	-0.04	101
	p-tau217 ~ YKL-40	0.49 [0.35-0.63]	0.39 [0.21-0.54]	-0.1	101
	p-tau181 ~ YKL-40	0.55 [0.41-0.67]	0.44 [0.28-0.59]	-0.11	101
	Total tau ~ YKL-40	0.53 [0.37-0.66]	0.45 [0.27-0.59]	-0.08	101
	p-tau205 ~ YKL-40	0.58 [0.43-0.7]	0.51 [0.35-0.65]	-0.07	101
	NfL ~ YKL-40	0.65 [0.49-0.76]	0.64 [0.49-0.75]	0	101
Aβ+ sMC	p-tau217 ~ p-tau181	0.95 [0.92-0.97]	0.96 [0.94-0.98]	0.01	107
	p-tau217 ~ Total tau	0.93 [0.89-0.95]	0.92 [0.86-0.95]	-0.01	107
	p-tau181 ~ Total tau	0.96 [0.93-0.97]	0.94 [0.9-0.96]	-0.02	107
	p-tau217 ~ p-tau205	0.92 [0.86-0.95]	0.9 [0.84-0.94]	-0.02	107
	p-tau181 ~ p-tau205	0.88 [0.81-0.92]	0.9 [0.83-0.94]	0.02	107
	Total tau ~ p-tau205	0.9 [0.83-0.94]	0.92 [0.87-0.95]	0.02	107
	p-tau217 ~ NfL	0.46 [0.28-0.63]	0.58 [0.42-0.72]	0.12	98
	p-tau181 ~ NfL	0.37 [0.17-0.55]	0.58 [0.41-0.72]	0.21	98
	Total tau ~ NfL	0.39 [0.2-0.57]	0.65 [0.49-0.77]	0.26	98
	p-tau205 ~ NfL	0.53 [0.35-0.68]	0.66 [0.5-0.78]	0.13	98
	p-tau217 ~ YKL-40	0.45 [0.28-0.61]	0.49 [0.3-0.63]	0.03	102
	p-tau181 ~ YKL-40	0.47 [0.29-0.62]	0.51 [0.34-0.65]	0.04	102
	Total tau ~ YKL-40	0.53 [0.36-0.67]	0.62 [0.47-0.74]	0.09	102
	p-tau205 ~ YKL-40	0.49 [0.31-0.65]	0.6 [0.45-0.72]	0.11	102
	NfL ~ YKL-40	0.45 [0.26-0.62]	0.68 [0.54-0.78]	0.23	99

Groups are defined as NC, amyloid-negative non-carriers; Aβ- MC, amyloid-negative mutation carriers; Aβ+ aMC, amyloid-positive asymptomatic mutation carriers with CDR = 0; and Aβ+ sMC, amyloid-positive symptomatic mutation carriers with CDR > 0.

Supplementary Table 6. Spearman correlations of CSF biomarkers (raw and Aβ40-normalized) with clinical, fluid and imaging outcomes in Aβ+ sMC.

Outcome	Predictor	ρ [95% CI]		Δρ	FDR-corrected	*n*
		Raw levels	Normalized to Aβ40		P_{shift}	
CDR Sum of Boxes	p-tau217	0.21 [0.01, 0.40]	0.46 [0.30, 0.61]	0.25	0.017*	107
	p-tau181	0.08 [-0.13, 0.28]	0.42 [0.25, 0.57]	0.34	<0.001***	107
	Total tau	0.16 [-0.05, 0.35]	0.56 [0.42, 0.68]	0.4	<0.001***	107
	p-tau205	0.33 [0.13, 0.50]	0.58 [0.44, 0.71]	0.26	0.008**	107
	NfL	0.76 [0.67, 0.83]	0.81 [0.74, 0.87]	0.05	0.279	99
	YKL-40	0.28 [0.10, 0.44]	0.58 [0.44, 0.71]	0.3	<0.001***	103
MMSE	p-tau217	-0.30 [-0.47, -0.12]	-0.45 [-0.60, -0.29]	-0.15	0.154	107
	p-tau181	-0.20 [-0.37, -0.02]	-0.42 [-0.57, -0.26]	-0.21	0.035*	107
	Total tau	-0.29 [-0.45, -0.12]	-0.54 [-0.67, -0.40]	-0.25	0.005**	107
	p-tau205	-0.40 [-0.54, -0.24]	-0.56 [-0.68, -0.42]	-0.16	0.06	107
	NfL	-0.66 [-0.77, -0.54]	-0.69 [-0.77, -0.58]	-0.03	0.808	99
	YKL-40	-0.33 [-0.48, -0.16]	-0.55 [-0.67, -0.41]	-0.22	0.026*	103
CSF NfL	p-tau217	0.46 [0.28, 0.63]	0.53 [0.36, 0.69]	0.07	0.559	98
	p-tau181	0.37 [0.17, 0.55]	0.48 [0.30, 0.65]	0.11	0.382	98
	Total tau	0.39 [0.20, 0.56]	0.49 [0.31, 0.66]	0.11	0.382	98
	p-tau205	0.52 [0.35, 0.68]	0.58 [0.42, 0.73]	0.06	0.601	98
	YKL-40	0.45 [0.26, 0.62]	0.48 [0.32, 0.62]	0.03	0.844	99
Plasma NfL	p-tau217	0.33 [0.14, 0.50]	0.56 [0.41, 0.69]	0.24	0.013*	99
	p-tau181	0.19 [-0.02, 0.37]	0.52 [0.37, 0.66]	0.34	<0.001***	99
	Total tau	0.22 [0.02, 0.40]	0.59 [0.45, 0.71]	0.37	<0.001***	99
	p-tau205	0.38 [0.19, 0.54]	0.61 [0.48, 0.74]	0.24	0.015*	99
	NfL	0.72 [0.61, 0.81]	0.78 [0.68, 0.86]	0.06	0.382	98
	YKL-40	0.30 [0.10, 0.47]	0.56 [0.40, 0.69]	0.26	0.011*	100
MRI composite	p-tau217	-0.29 [-0.47, -0.09]	-0.50 [-0.66, -0.31]	-0.22	0.049*	97
	p-tau181	-0.15 [-0.35, 0.05]	-0.47 [-0.64, -0.29]	-0.32	0.005**	97
	Total tau	-0.23 [-0.42, -0.04]	-0.60 [-0.73, -0.42]	-0.37	<0.001***	97
	p-tau205	-0.37 [-0.55, -0.18]	-0.59 [-0.73, -0.42]	-0.22	0.030*	97
	NfL	-0.63 [-0.75, -0.48]	-0.74 [-0.82, -0.63]	-0.11	0.184	89
	YKL-40	-0.23 [-0.42, -0.03]	-0.54 [-0.68, -0.38]	-0.31	<0.001***	93
FDG-PET composite	p-tau217	-0.17 [-0.36, 0.03]	-0.38 [-0.54, -0.20]	-0.21	0.072	91
	p-tau181	-0.07 [-0.28, 0.14]	-0.37 [-0.53, -0.19]	-0.3	0.008**	91
	Total tau	-0.15 [-0.35, 0.05]	-0.50 [-0.65, -0.33]	-0.35	<0.001***	91

	p-tau205	-0.27 [-0.46, -0.06]	-0.47 [-0.63, -0.30]	-0.2	0.065	91
	NfL	-0.58 [-0.71, -0.43]	-0.68 [-0.79, -0.55]	-0.1	0.228	84
	YKL-40	-0.18 [-0.37, 0.02]	-0.48 [-0.64, -0.30]	-0.31	0.013*	88
Tau PET composite	p-tau217	0.39 [-0.61, 1.00]	0.53 [-0.35, 1.00]	0.14	0.845	7
	p-tau181	0.53 [-0.35, 1.00]	0.56 [-0.21, 1.00]	0.07	0.909	7
	Total tau	0.33 [-0.62, 0.96]	0.53 [-0.35, 1.00]	0.18	0.808	7
	p-tau205	0.39 [-0.61, 1.00]	0.53 [-0.35, 1.00]	0.14	0.845	7
	NfL	-0.17 [-1.00, 0.76]	0.19 [-1.00, 0.75]	0.46	0.560	7
	YKL-40	0.04 [-0.92, 0.96]	0.06 [-0.65, 1.00]	-0.04	0.971	7
Global PiB summary score	p-tau217	0.03 [-0.19, 0.26]	0.10 [-0.13, 0.33]	0.07	0.627	92
	p-tau181	-0.00 [-0.22, 0.23]	0.09 [-0.15, 0.32]	0.09	0.548	92
	Total tau	0.07 [-0.15, 0.28]	0.19 [-0.04, 0.40]	0.12	0.382	92
	p-tau205	0.08 [-0.12, 0.29]	0.17 [-0.05, 0.39]	0.09	0.548	92
	NfL	-0.08 [-0.31, 0.15]	-0.00 [-0.24, 0.23]	0.07	0.627	83
	YKL-40	0.17 [-0.04, 0.37]	0.26 [0.02, 0.48]	0.09	0.501	87

Shown here are Spearman correlations of CSF biomarkers with clinical outcomes, without and with normalization to Aβ40, and the Δρ (Δρ = ρ(normalized biomarker) - ρ(raw biomarker)). Models are bootstrapped (n = 2,000), and p-values are shown for a significant difference between correlations with and without normalization (P_{shift}). P-values reflect the extremeness of the mean Aβ40-normalized ρ relative to the raw-ρ bootstrap distribution (two-sided tail probability), not a direct hypothesis test of Δρ=0. Abbreviations: Aβ = amyloid-beta; CDR= Clinical Dementia Rating; FDG = fluorodeoxyglucose; FDR = false discovery rate; MRI = magnetic resonance imaging; NfL = neurofilament light; PiB = Pittsburgh compound B; YKL-40 = chitinase-3-like protein 1.